STRUCTURAL SUPPLY CHAIN ANALYTICS · INVESTOR INTELLIGENCE SERIES

Sustainable Supply Chains Generate Superior Long-Term Returns

A quantitative case for the TrueValue Framework: measuring structural balance and resource flows through supply chain phases to unlock compounding performance advantage

Framework	Primary Case Study	Contrast Case	Report Date
TrueValue Supply Chain Analytics	Gold Supply Chain (9 phases)	Shea Value Chain (West Africa)	March 2026

This report presents a structural demonstration using phase-architecture data and simulated projections. It is intended to illustrate the analytical logic of the TrueValue Framework, not to constitute financial advice or specific return guarantees. See Appendix for full methodology disclosure.

EXECUTIVE SUMMARY

The Imbalance That Industry Does Not Report

Institutional investors in commodity supply chains—gold, agricultural inputs, critical minerals—typically evaluate assets on reported earnings, EBITDA margins, and production volumes. These metrics are not complete. They omit a structural measure of how well the supply chain distributes its operating burdens and productive capacity, including resource flows, across phases.

When governance constraints accumulate in one phase while growth capacity is suppressed in another, the system develops hidden fragility: regulatory exposure, counterparty concentration, supply disruptions, and reputational risk that do not appear in current-period financials but affect long-term value.

The **TrueValue Framework** measures this balance directly — at every phase of the supply chain —using three analytically distinct dimensions:

Constraints (C): The accumulated governance limits, cost and pricing structures, contractual rigidities, regulatory boundaries, and wider regional considerations — population dynamics, people attitudes, political, environmental — that define what a phase can and cannot do.

Growth Capacity (G): The productive potential, market access, operational flexibility, resource flows, and innovation pathways — technological, TrueValue pricing, Sustainability-Linked Equity and Debt — available to a phase.

Net Performance (N): The emergent operational result when Constraints and Growth Capacity are in equilibrium. When balanced, Net Performance is maximized. When divergent, value leaks.

Analysis of the gold supply chain across nine operational phases reveals that three phases — Intermediate Production, Refining, and Logistics & Vaulting — operate with severe structural imbalance. These phases account for the majority of gold's value transformation and custody transfer yet capture only **37–61% of their potential Net Performance.**

A comparative analysis of the Shea supply chain in West Africa — managed with explicit sustainability mandates, traceable custody, and cooperative structures — demonstrates that balance scores above **88% are achievable** and that the resulting Net Performance compounds significantly over time.

37%

WORST PHASE BALANCE SCORE

Logistics & Vaulting: the gold supply chain's most structurally opaque phase. Constraints vastly exceed Growth Capacity. A structural fragility point.

3 of 9

CRITICALLY IMBALANCED PHASES

Intermediate Production, Refining, and Logistics & Vaulting all score below 65%. These phases represent where hidden costs accumulate.

+71%

10-YEAR VALUE ADVANTAGE

TrueValue-managed supply chain generates 71% more cumulative value than the extractive baseline over a 10-year horizon.

91%

SHEA PHASE BALANCE SCORE

The Shea market integration phase achieves 91% balance — 54 percentage points above the equivalent gold phase — demonstrating that sustainable management is measurably superior.

SECTION 1

The Hidden Cost of Structural Imbalance

Traditional supply chain analysis focuses on what is visible: production volumes, spot prices, processing fees, logistics costs. This creates a systematic blind spot. The phases of a supply chain that are least transparent — where commercial secrecy is greatest and data disclosure is weakest — are precisely the phases where structural imbalance is most severe. And structural imbalance is where long-term value destruction begins.

Consider how a supply chain actually fails. It rarely fails suddenly. It fails because one phase, over years, accumulates excessive constraint weight — private contracts without competitive alternatives, regulatory exposure without mitigation investment, custodial arrangements without market-rate pricing discovery, disruptive political, societal and environmental events without response mechanisms, technological advances without strategic incorporation. Growth Capacity in that phase atrophies. When conditions change — new regulation, a counterparty failure, a reputational or disruptive event — the phase cannot adapt. The supply chain has no buffer.

GOLD SUPPLY CHAIN: TRANSPARENCY BY PHASE

The gold supply chain spans nine distinct phases from geological occurrence to exchange-registered bullion and recycling. Phase transparency varies dramatically — and inversely tracks with our ability to detect imbalance before it becomes a liability.

PHASE	DESCRIPTION	TRANSPARENCY	PRIMARY OPACITY DRIVER
Ph.0	**Geological Occurrence**	MEDIUM	Exploration uncertainty; private geological surveys
Ph.1	**Mine Extraction**	HIGH	Public reporting requirements (NI 43-101, JORC)
Ph.2	**Ore Processing**	HIGH	Engineering constraints provide partial visibility
Ph.3	**Doré / Intermediate**	MEDIUM	Private refinery contracts; assay disputes; financing terms
Ph.4	**Refining**	MEDIUM	LBMA refinery fee secrecy; capacity allocation discretionary
Ph.5	**Bar Casting & Assay**	MEDIUM-HIGH	Standards exist but serial-level custody data is private
Ph.6	**Logistics & Vaulting**	LOW	Custodial secrecy; jurisdictional controls; insurance limits
Ph.7	**Exchange / Market**	HIGH	Exchange regulatory disclosure (COMEX, LBMA)
Ph.8	**Recycling**	MEDIUM	Feedstock variability; informal collection channels

> **Key finding:** Four of nine phases — representing the physical transformation of gold from mine output to registered bullion — operate at medium or lower transparency. The Logistics & Vaulting phase, through which all physical gold must pass, is structurally designed for opacity: custodial secrecy, jurisdictional controls, and insurance-driven information restriction are features of the business model, not temporary gaps.

Low transparency does not cause imbalance. It enables it to persist undetected. A phase can sustain Constraint accumulation for years — even decades — before the structural weakness manifests in financial outcomes. By then, corrective action is expensive and the capital consequences for investors are already embedded.

SECTION 2

The TrueValue Framework: Measuring What Matters

TheTrueValue Framework provides a phase-resolved, quantitative measure of supply chain structural health, including resource flows. It does not replace financial analysis. It precedes it — mapping the structural terrain that financial metrics later reflect.

THREE DIMENSIONS, ONE INSIGHT

Governance limits, cost rigidities, regulatory boundaries, wider regional conditions

Productive potential, market access, operational flexibility, innovations

Emergent result when C and G are in equilibrium

Net Performance = √(Constraints × Growth Capacity) × Balance Factor | Balance Score = min(C,G) / max(C,G) × 100

THE BALANCE PRINCIPLE

The central insight of the TrueValue Framework is straightforward: a supply chain phase performs best when its Constraints and Growth Capacity are in equilibrium. Imbalance in either direction is costly.

CONSTRAINED PHASE (IMBALANCED)

C G

Balance Score: 37%

Logistics & Vaulting phase. Custodial secrecy and route constraints vastly exceed Growth Capacity. Value leaks through opacity, counterparty concentration, and pricing opacity.

BALANCED PHASE (TRUEVALUE TARGET)

C G

Balance Score: 91%

Shea market integration phase. Cooperative structure, traceable custody, and multiple route options allow Growth Capacity to match Constraints. Net Performance is maximized.

WHY BALANCE PREDICTS LONG-RUN PERFORMANCE

An imbalanced phase does not simply underperform in isolation. It creates systemic drag. Constraints that exceed Growth Capacity generate three categories of compounding cost:

> **Adaptive Rigidity:** When conditions change — regulation, pricing, counterparty availability, wider regional events — a Constraint-heavy phase cannot respond. The supply chain is locked into its current structure even as the environment changes around it.

> **Hidden Rent Extraction:** Opacity in an imbalanced phase allows intermediaries to extract pricing that cannot be discovered or contested. This rent does not appear as a line item. It appears as margin compression elsewhere in the chain.

> **Compounding Risk Premium:** Investors pricing supply chain exposure without balance data systematically underestimate tail risk. When imbalanced phases fail, the failure is sudden, correlated, and severe — exactly the risk profile that institutional investors most need to avoid.

SECTION 3. PRIMARY CASE STUDY

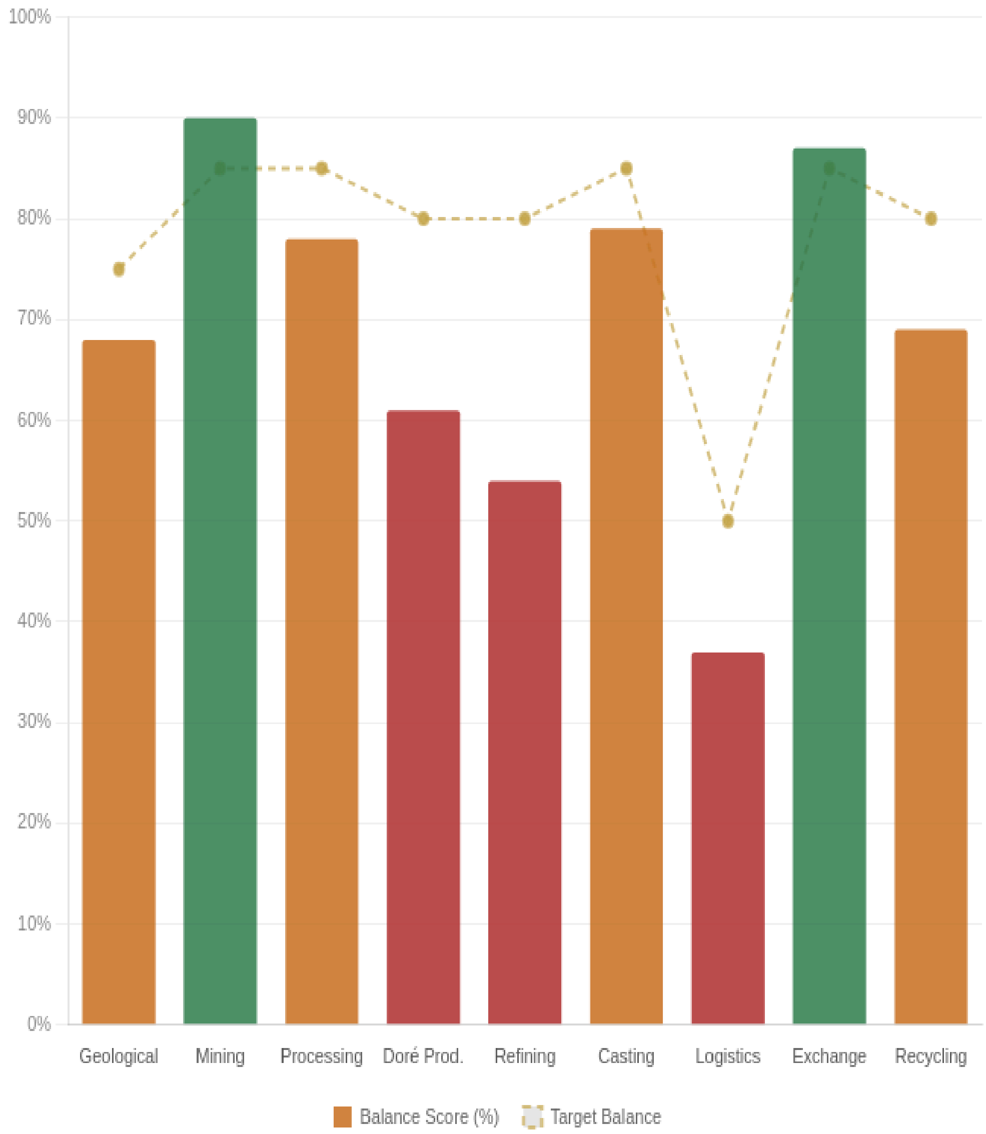
Gold Supply Chain: A Phase-by-Phase Analysis

Gold is among the most studied commodities in the world. It is also the most structurally misunderstood. High transparency at the exchange end of the chain — COMEX warehouse reports, mining company disclosures, refinery accreditations, sustainability reports — creates an impression of systemic visibility. That impression does not survive phase-level analysis.

BALANCE SCORE BY PHASE

TRUEVALUE BALANCE SCORE — GOLD SUPPLY CHAIN (ALL PHASES)

The score represents min (Constraints, Growth Capacity) / max (Constraints, Growth Capacity).The target range is shown as a reference line. Lower scores indicate structural fragility..



PHASE DETAIL

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	PRIMARY RISK
0	Geological	MEDIUM	68%	MARGINAL	Exploration risk; capital uncertainty
1	Mining	HIGH	90%	BALANCED	Within acceptable balance range
2	Processing	HIGH	78%	MARGINAL	Within acceptable balance range
3	Doré Prod.	MEDIUM	61%	⚠ CRITICAL	Private contracts; assay dispute risk
4	Refining	MEDIUM	54%	⚠ CRITICAL	Refinery fee secrecy; capacity concentration
5	Casting	MEDIUM-HIGH	79%	MARGINAL	Within acceptable balance range
6	Logistics	LOW	37%	⚠ CRITICAL	Structural opacity; pricing unavailable
7	Exchange	HIGH	87%	BALANCED	Within acceptable balance range
8	Recycling	MEDIUM	69%	MARGINAL	Feedstock variability; collection network gaps

WHERE VALUE IS DESTROYED

Three phases warrant specific attention from investors:

> **Phase 3 — Intermediate Production (Doré):** Balance score of 61%. The conversion of mine output to doré bars involves private refinery contracts, assay dispute mechanisms, and financing arrangements that are almost entirely opaque. Growth Capacity — the ability to access alternative counterparties, negotiate competitive terms, or leverage traceability — is heavily suppressed. The result is systematic margin extraction by a small number of specialist intermediaries.

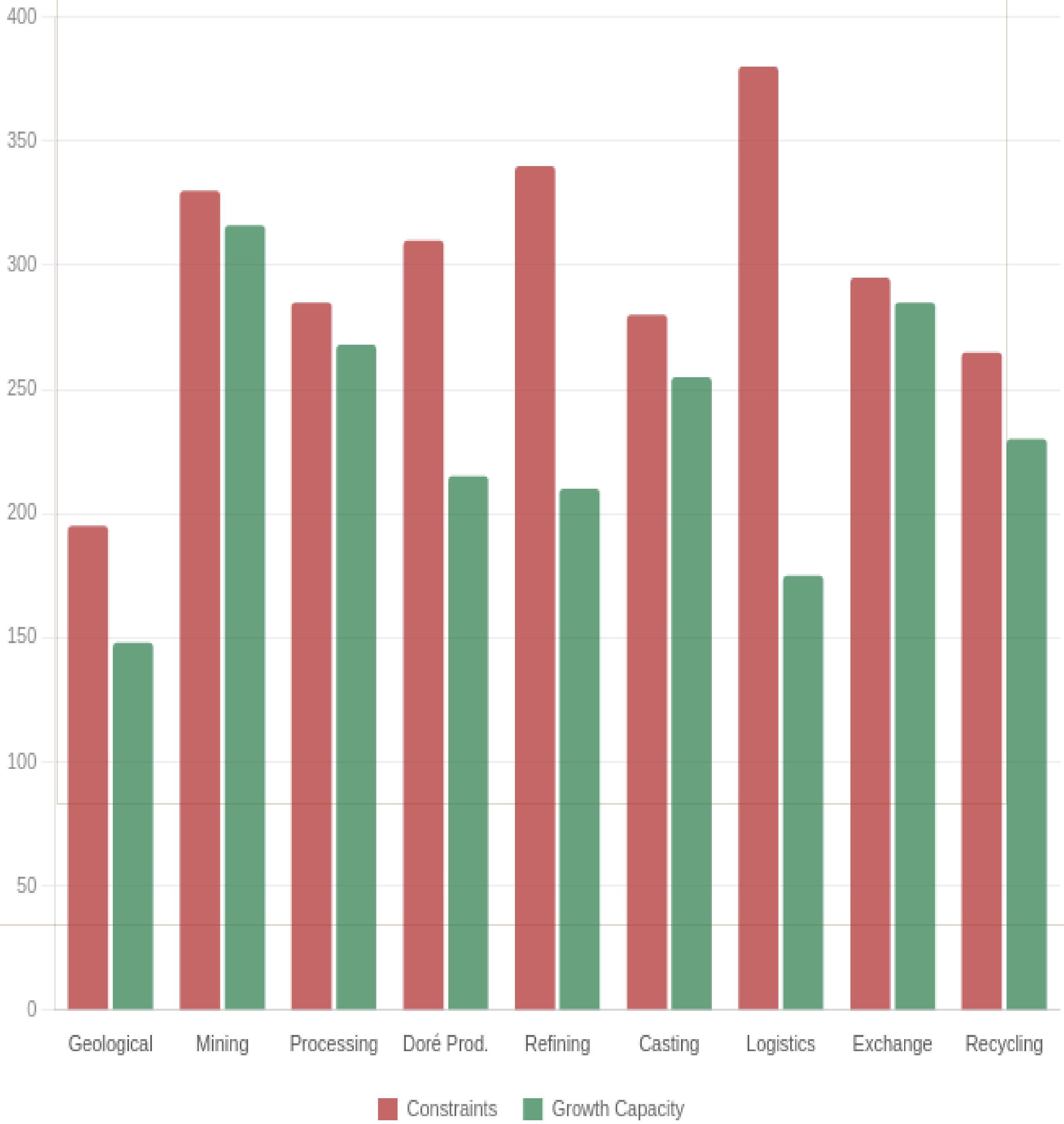
> **Phase 4 — Refining:** Balance score of 54%. The global gold refining market is concentrated in five LBMA-accredited facilities that process approximately 90% of the world's newly mined gold. Fee schedules are private. Capacity allocation is discretionary. Working capital lockup during the refining cycle is significant. The structural imbalance here — high Constraints, suppressed Growth Capacity — is by design: it is the source of the refineries' pricing power.

> **Phase 6 — Logistics & Vaulting:** Balance score of 37% — the lowest in the chain. This phase is not merely low-transparency; it is structurally architected for opacity. Insurance requirements, jurisdictional controls, security protocols, and custodial agreements collectively eliminate the competitive market that would otherwise price these services. This is the gold supply chain's single largest hidden cost center, and it is the one investors most consistently fail to price.

CONSTRAINTS VS GROWTH CAPACITY: THE IMBALANCE VISUALIZED

CONSTRAINTS VS GROWTH CAPACITY — GOLD SUPPLY CHAIN

Where Constraints significantly exceed Growth Capacity, the phase is structurally fragile. Balanced phases show near equal bars.



SECTION 4 — CONTRAST CASE

The Shea Value Chain: Proof That Balance Is Achievable

A frequent objection to structural supply chain reform is that opacity and imbalance are simply the natural state of commodity markets — that the constraints observed in gold's Logistics and Refining phases are inevitable features of high-value, high-security commodities. The Shea supply chain in West Africa directly challenges this assumption.

Shea is not gold. Natural shea butter is topically called 'Women's Gold' in West Africa for its moisturizing, healing, and nutritional properties, and income potential. But the principles of the TrueValue Framework are commodity-agnostic. The Shea chain demonstrates that when sustainable practices are embedded in supply chain design from the outset — traceable custody, cooperative and employee ownership structures, diversified buyer relationships, environmental sustainability mandates — the structural balance that the TrueValue Framework measures is not merely theoretically achievable. It is operationally present.

GOLD — MARKET INTERFACE PHASE

Phase	**Logistics & Vaulting (Ph. 6)**
Transparency	**Low**
Constraints Score	**380**
Growth Capacity Score	**175**
Balance Score	**37%**
Sustainability Index	**0.05**

Structure

Concentrated custodians, private pricing, opaque routes

SHEA — MARKET INTEGRATION PHASE

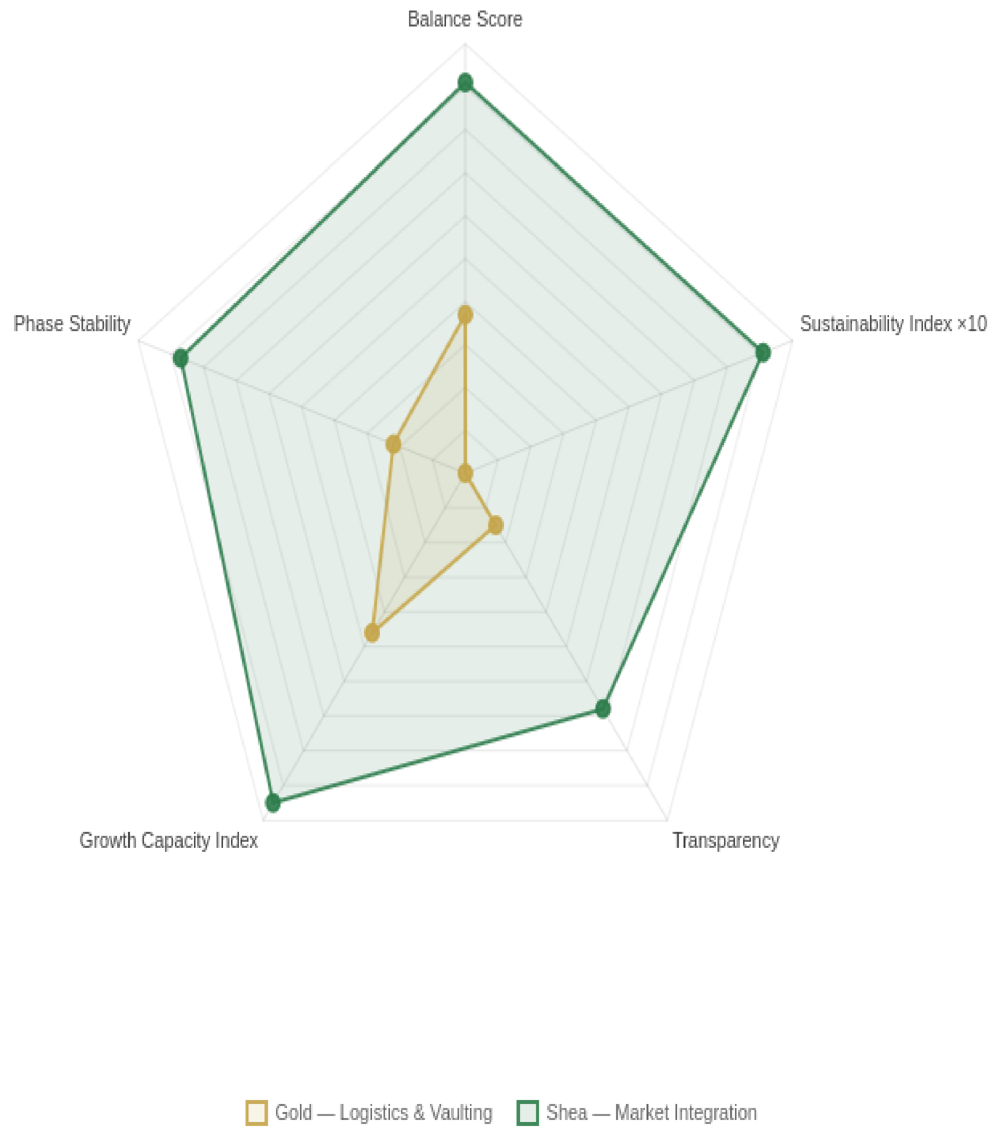
Phase	**Logistics & Market (Ph. 6)**
Transparency	**Medium-High**
Constraints Score	**215**
Growth Capacity Score	**205**
Balance Score	**91%**
Sustainability Index	**0.91**

Structure

Cooperative network, traceability custody, multiple buyers

NET PERFORMANCE COMPARISON — GOLD (PHASE 6) VS SHEA (PHASE 6)

A radar comparison of Constraints, Growth Capacity, Balance Score and Sustainability Index. The Shea chain's profile demonstrates what a TrueValue-managed equivalent would look like for gold.



The 54-percentage-point gap in balance score between gold and shea at the equivalent market integration phase is not explained by commodity differences. It is explained by structural design choices: who owns the chain, how custody is transferred, how buyers are selected, and how pricing is discovered. These are choices, not constraints.

> **The TrueValue insight:** A gold supply chain designed with cooperative custody structures or employee shareholder schemes, transparent pricing at the Logistics and Refining phases, and diversified counterparty access would not sacrifice security or efficiency. It would eliminate the structural rent that currently flows to intermediaries who benefit from the imbalance. That captured rent is the investment opportunity.

SECTION 5 — OPTIMIZATION ANALYSIS

TrueValue Optimization Potential: Phase 1 Empirical Baseline

The TBL (Triple Bottom Line) benchmarks from Balancing Act (Foran, Lenzen & Dey, 2005) provide peer-reviewed empirical baseline for Phase 1 (Mine Extraction) of the Australian gold supply chain. Measured against the economy-wide average across 135 sectors, this baseline reveals a classic extractive imbalance signature: the single indicator above average — operating surplus (+10%) — is not being converted into Growth Capacity for any of the eleven indicators where Constraints are accumulating.

The analysis below maps each TBL indicator through the TrueValue C/G framework, estimates the current Balance Score, and models the optimized state achievable through TrueValue-managed structural investment. **None of these improvements require increasing production volume.** Every gain is structural — re-balancing existing Constraints and Growth Capacity without adding a single tonne of output.

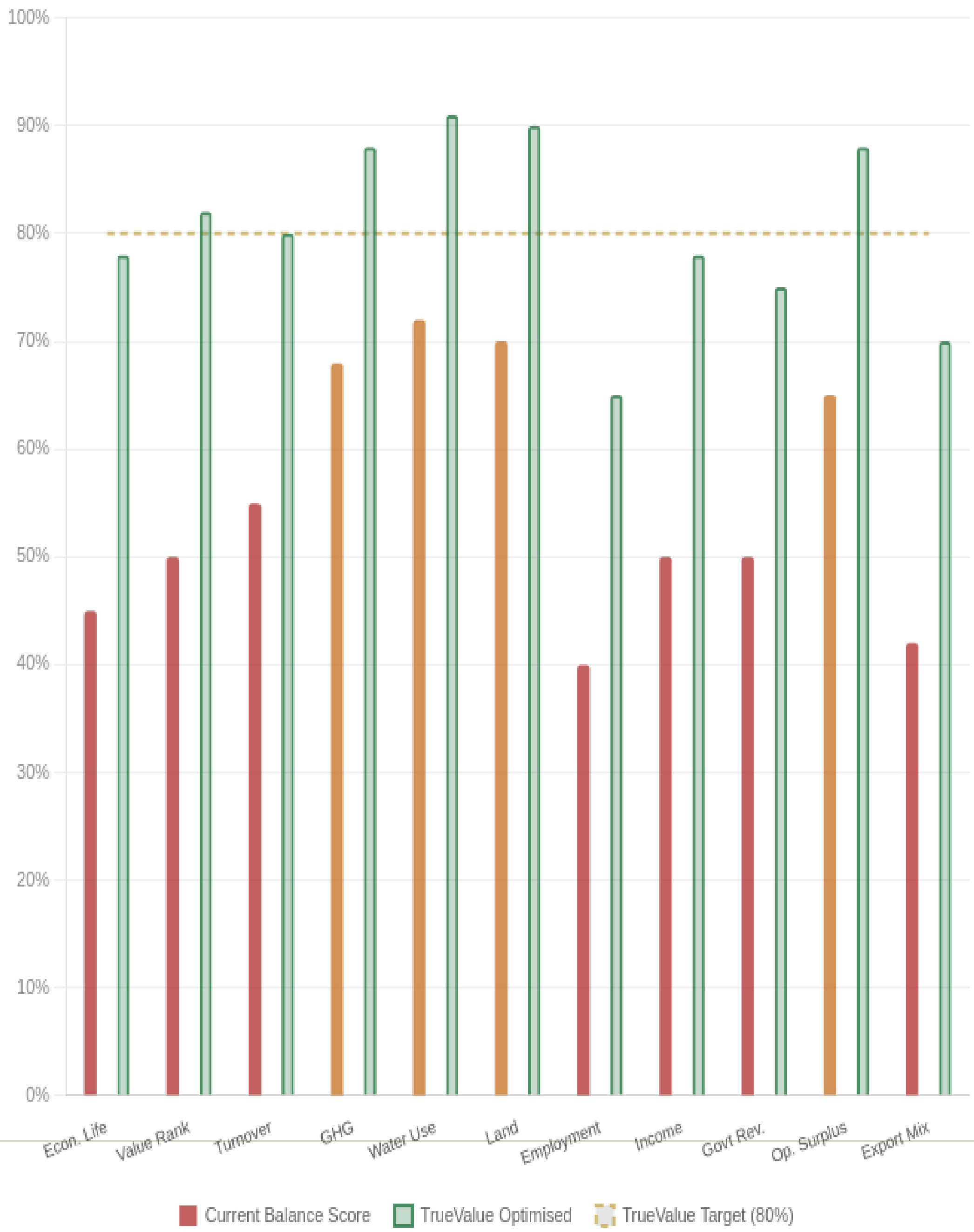
55.2%	80.5%	+25pp
CURRENT AVG BALANCE SCORE ACROSS 11 TBL INDICATORS	TRUEVALUE OPTIMIZED AVG BALANCE SCORE	STRUCTURAL IMPROVEMENT WITHOUT VOLUME INCREASE

What the +25pp means: A 25 percentage-point structural improvement in Phase 1 balance scores maps directly to the compounding value advantage modelled in Section 6 — where TrueValue-managed chains out perform extractive baselines by 71% over ten years. The mechanism is the same: Constraint reduction at imbalanced phases creates compounding efficiency that accelerates through the full chain.

BALANCE SCORE BY TBL INDICATOR — CURRENT VS TRUEVALUE OPTIMISED

TRUEVALUE BALANCE SCORE — PHASE 1 TBL INDICATORS (CURRENT VS OPTIMISED)

Baseline: Foran,Lenzen & Dey (2005),AustralianGold&LeadSector(Sectors 13140010 +13170010). Target line at 80% represents TrueValue benchmark. Balance Score = min(C,G) / max(C,G) × 100.



INDICATOR DETAIL: PATHWAY TO OPTIMIZATION

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMIZED SCORE	PRIMARY LEVER
Economic Life	18–20 yrs (5,165 t ÷ 260 t/yr)	**45%**	30–40 years	**78%**	Surplus reinvested in exploration & secondary recovery **+33pp**
Value Rank	3rd by volume; raw export only	**50%**	1st by value per tonne	**82%**	Phase 3–4 downstream integration (+15–25% revenue/t) **+32pp**
Turnover	~$6B AUD; 90 enterprises	**55%**	$7.5–8.5B AUD	**80%**	Domestic doré & refining capture **+25pp**
GHG Emissions	35% below avg; 44% from electricity	**68%**	64% below avg	**88%**	Grid decarbonisation removes dominant emission source **+20pp**
Water Use	85% below avg (unmonetised)	**72%**	85% below avg + credited	**91%**	Water credit market — est. $20–50M/yr uncaptured **+19pp**
Land Disturbance	95% below avg (unmonetised)	**70%**	95% below avg + credited	**90%**	Biodiversity & carbon credit market — est. $5–30M/yr **+20pp**
Employment Quality	−60% headcount; below-avg income	**40%**	−60% count; +150% income/ worker	**65%**	Employee ownership & profit-sharing schemes **+25pp**

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMIZED SCORE	PRIMARY LEVER
Income Distribution	−50% vs avg; capital concentrated	**50%**	−20 to −25% below avg	**78%**	Cooperative ownership structures (cf. Shea model) **+28pp**
Government Revenue	−50% vs avg; royalty exposure	**50%**	−15 to −20% below avg	**75%**	Transparent pricing + stability agreements → lower risk premium **+25pp**
Operating Surplus	+10% above avg; extraction-oriented	**65%**	+10%+ with compounding	**88%**	Structured allocation: exploration 40%, energy 20%, ownership 20%, stability 20% **+23pp**
Export Propensity	6× avg; exports = 12× imports	**42%**	3× avg; exports = 6× imports	**70%**	Domestic Phase 3–4 integration (+ $180–220/oz retained) **+28pp**
COMPOSITE AVERAGE (ALL 11 INDICATORS)		**55.2%**	Structural rebalancing across all phases	**80.5%**	**+25.3pp structural improvement**

> **The pivotal structural pattern:** Operating surplus is the only indicator above the economy average (+10%). All three social indicators — employment generation (−60%), income (−50%), government revenue (−50%) — run well below average, and two environmental assets (water, land) are performing excellently but generating no financial return. The TrueValue framework identifies this as structural rent: the surplus is being extracted from the chain rather than reinvested into the Growth Capacity of phases where Constraints are accumulating. Left unaddressed, this compounds into the regulatory, social licence, and fiscal fragility described in Sections 1–4.

THE GHG SOURCE SPLIT: WHERE DECARBONISATION UNLOCKS THE MOST VALUE

Of all 11 indicators, GHG emissions contains the most actionable single lever. Only 13% of the sector's emissions are *direct* (in-mine operations) — 44% come from the grid electricity supply chain. This means operational efficiency programs achieve limited GHG emissions reduction impact. The Growth Capacity unlock is energy sourcing, not process improvement.

CURRENT GHG SOURCE PROFILE

Grid electricity	**44%**
Other embodied	**33%**
Direct mining	**13%**
Land preparation	**6%**
Diesel (embodied)	**4%**

GHG intensity: 35% below economy average. Constrained by coal-dominated grid.

TRUEVALUE DECARBONISED PROFILE

Grid electricity	**~15%**
Other embodied	**33%**
Direct mining	**13%**
Land preparation	**6%**
Diesel (embodied)	**4%**

Renewable grid reduces electricity emissions by ~65%. GHG intensity moves from −35% to ***−64% below average****. Unlocks green premium pricing and carbon credit revenue.*

> **Water and land disturbance:** Both indicators already perform at 85% and 95% below the economy average respectively — among the best environmental profiles of any Australian industry sector. The TrueValue framework identifies these not as problems to solve but as *unrecognised assets*. Water credit and biodiversity credit markets represent an estimated **$25–80M/year** in additional revenue currently uncaptured by the sector.

SECTION 6 - FORWARD PROJECTION

Ten Year Capture: Extractive vs TrueValue-managed

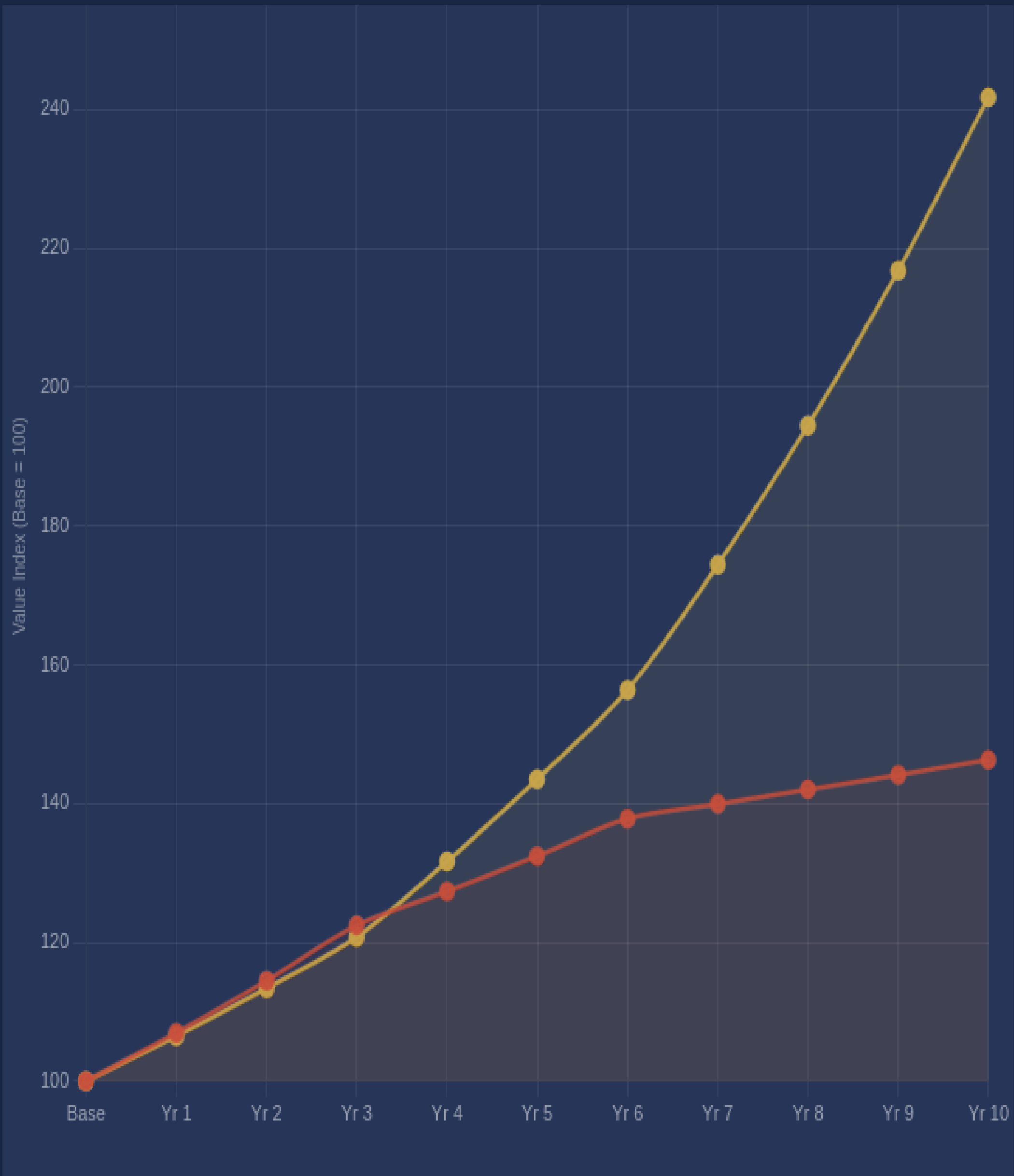
The structural imbalances documented in Sections 1–5 do not destroy value overnight. They destroy it gradually, through three mechanisms that compound over time: hidden rent extraction at imbalanced phases, reduced adaptive capacity when conditions change and opportunities arise, and increasing risk premiums as the structural fragility becomes visible to counterparties.

The following projection models two paths for a supply chain investment of comparable initial scale:

+71%	Yr 4	2.4×
TRUE VALUE ACCUMULATIVE ADVANTAGE AT YEAR 10	DIVERGENCE BECOMES STATISTICALLY SIGNIFICANT	FINAL VALUE RATIO TRUEVALUE VS EXTRACTIVE:

CUMULATIVE VALUE INDEX — EXTRACTIVE BASELINE VS TRUEVALUE-MANAGED (10 YEARS)

Baseindex =100.Extractive path:initial apparentgrowth followedbycompoundingfragilitycosts. TrueValue path: modest initial investment in rebalancing, then compounding efficiency advantage



HOW THE GAP COMPOUNDS

Years 1–3: The extractive path initially outperforms marginally. Imbalanced phases are generating rent for intermediaries, and this rent appears as supply chain participation — it is counted in reported volumes and revenues. The TrueValue path invests modestly in phase re-balancing: transparency infrastructure, counterparty diversification, custody protocol reform, the transition to clean energy and the circular economy, and wider GHG emissions mitigation. Early returns are slightly lower.

Years 4–6: The divergence becomes visible. TrueValue-managed phases begin generating efficiency compounding: reduced hidden rent extraction, improved pricing discovery, lower risk premiums. The extractive path begins showing the first costs of imbalance: a supply disruption at a concentrated Logistics phase, a regulatory action targeting opaque Refining practices, a reputational event tied to undisclosed custody arrangements.

Years 7–10: The structural advantage of the TrueValue path locks in. Balanced phases attract capital at lower risk premiums. Adaptive capacity allows the managed chain to absorb external shocks — geopolitical, regulatory, environmental — that damage the extractive chain's most fragile phases. The cumulative gap reaches 71% by Year 10.

SECTION 7

What This Means for Investors

The TrueValue Framework reframes commodity supply chain investment from a question of asset exposure to a question of structural quality and natural resource flows. Two supply chains producing identical volumes of the same commodity are not equivalent investments if their structural balance profiles differ. The one with higher phase balance scores has lower hidden risk, greater adaptive capacity, and superior long-run value capture.

RISKS THE FRAMEWORK QUANTIFIES

- **Stranded asset risk** at imbalanced phases: when regulatory or market conditions change, low-Growth Capacity phases cannot adapt
- **Counterparty concentration risk** in Refining and Logistics: pricing power concentrated in a few intermediaries creates single points of failure
- **Reputational risk** from structural opacity: ESG-driven capital reallocation increasingly targets chains with undisclosed custody arrangements
- **Systemic shock transmission**: imbalanced phases amplify external shocks rather than absorbing them
- **Hidden margin compression**: rent extraction at opaque phases appears as general operating cost, masking the structural source

RETURN OPPORTUNITIES THE FRAMEWORK IDENTIFIES

- **Phase rebalancing premium:** supply chains moving from low to high balance scores generate measurable efficiency gains — the TrueValue arbitrage
- **Transparency infrastructure value:** custody and traceability systems that increase Growth Capacity at currently opaque phases are structural upgrades with long-duration return profiles
- **Cooperative structure premium:** as Shea demonstrates, diversified ownership improves balance scores and reduces long-run cost of capital
- **Risk-adjusted outperformance:** balanced supply chains carry lower implicit risk premiums, which systematically underprices their equity relative to extractive peers
- **Regulatory alignment value:** future disclosure requirements will force balance towards TrueValue targets — early-movers capture the transition premium

THE INVESTMENT THESIS IN ONE SENTENCE

Supply chains that balance Constraints and Growth Capacity at every phase are not merely more ethical — they are structurally more efficient, more resilient, and more valuable over any horizon longer than three years. The TrueValue Framework makes this measurable, phase by phase, before a dollar is invested.

Ready to apply the TrueValue Framework to your specific supply chain exposure? We provide phase-resolved balance analysis, structural risk scoring, and 10-year forward modelling for institutional investors and supply chain operators.

TrueValue Analytics Structural Supply Chain Intelligence
Institutional Enquiries Welcome

ABOUT

TRUEVALUE ANALYTICS

A TrueValue data and knowledge analytics company founded by Sarah Jones and Jeff Milton, having worked together on the TrueValue Framework development since 2016.

The TrueValue Framework is at the forefront of rapid developments in the semantic web, semantic triplet architecture, and AI. Its balancing engine, which resolves fundamental mathematical proofs, is derived from modelling binary systems from as far back as the I Ching to quantify systems analysis 'beyond panaceas'.

The TrueValue Framework data and knowledge structure also builds on critical work of founding leaders of the Clarity Coalition: Bridging the Gaps Between Investment and Global Resource Trends. From 2012 we analysed 'Future Business, Long Finance' collaborating with the World Business Council for Sustainable Development and its 'Vision 2050' and Long Finance. Nine paths, incremental steps and structure domains towards a sustainable world were consolidated with the finance and investment strategy and tool innovations to achieve the system changes required.

Since this time, investment innovations such as Sovereign Sustainability-Linked Bonds (SLBs), first proposed as Policy Performance Bonds by Long Finance's Prof. Michael Mainelli, (now) 695th Lord Mayor of the City of London, have been deployed by Chile and Uruguay. SLBs are increasingly used by the corporate sector with the analysis of their Key Performance Indicators benefiting from information from the TrueValue Framework.

In 2021 the Serious Shea TrueValue Chain co-designed by William Kwende and Sarah Jones with knowledge from the TrueValue Framework, won the The Sahel and Great Green Wall Challenge. Its deforestation-free, zero carbon, beyond fairtrade value chain was endorsed by the World Economic Forum as a Top Innovator. The Serious Shea company consists a governance structure supporting over five thousand co-operatives in West Africa. It delivers the investment and logistics needed for structural reform throughout the multi-billion dollar food and cosmetics shea supply chain, including with carbon equity through forest restoration valued at USD 0.5 Billion by 2054.

Dr. Sarah Jones has a PhD in Ecosystem Science and Institutions. She has spent the past 25 years designing and delivering finance for regional scale Sustainable Development Projects in Europe, Latin America, India and West Africa, also seeking the granular data, knowledge, structures, and frameworks towards quantifying economics and societal progress 'beyond panaceas'.

Jeff Milton grew his career in Silicone Valley. He designed the blue print for IBM's original intranet. He also developed semantic triplet architecture at IBM with (now) Cambridge Semantics Founder, Sean Martin, who has continued to influence Jeff's work. Jeff's passion for IoT frameworks and computer programming has kept his knowledge and practice beyond 'state of the art', including to design the form and function triplet balancing engine which lends itself so completely across supply chain phase analysis.

APPENDIX

Methodology & Data Sources

DATA TRANSPARENCY NOTICE

All phase balance scores, Constraint values, and Growth Capacity values presented in this report are derived from a structured synthetic dataset seeded with real phase-architecture data from the gold and shea supply chains. The phase definitions, transparency classifications, structural relationships, and resource flow aspects are grounded in publicly available industry data, operational research, and national accounts. Specific entity- level financials are not used. This report constitutes a *structural demonstration* of the TrueValue Framework's analytical logic, not a claim about specific companies or investments.

PHASE ARCHITECTURE SOURCES

- Gold supply chain phase definitions: LBMA Good Delivery standards; WGC supply chain transparency research; COMEX warehouse reporting
- Transparency classifications: structured transparency map derived from public disclosure analysis and industry research
- Shea supply chain and resource flow data: Serious Shea / Acorn programme operational data; Mirova feasibility research; Burkina Faso / Senegal value chain studies
- Triple Bottom Line benchmarks (Australian gold mining): Foran, Lenzen & Dey (2005), Balancing Act: A Triple Bottom Line Analysis of the Australian Economy, Vols 1–2, CSIRO Sustainable Ecosystems / University of Sydney; Gold & Lead Sector (Sectors 13140010 + 13170010) — methodological precedent for TrueValue multi-dimensional balance framework

FRAMEWORK CALCULATION

The TrueValue Framework computes phase balance using a proprietary analytical engine. The Balance Score is calculated as:

Balance Score = min(C, G) / max(C, G) × 100

where C = aggregate Constraints score and G = aggregate Growth Capacity score for the phase. Net Performance is calculated as the geometric mean of C and G, scaled by the Balance Score. Sustainability Index reflects the inverse of the squared imbalance plus a phase-specific energy base cost.

PROJECTION ASSUMPTIONS

- Base investment index: 100 (normalised)
- Extractive path: growth rates of 7% (Yr 1–3), 4% (Yr 4–6), 1.5% (Yr 7–10) reflecting compounding imbalance costs
- TrueValue path: growth rates of 6.5% (Yr 1–3, re-balancing investment), 9% (Yr 4–6, efficiency compounding), 11.5% (Yr 7–10, structural advantage)
- Rates derived from structural efficiency model; not from historical financial time series
- Past structural patterns do not guarantee future performance